FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-023 ShopPulse Incremental Landing

Integrated Mini-Lab | restart-safe ingestion

Artifact	C3_023_SHOPPULSE_INCREMENTAL_LANDING_MINI_LAB_RC1
Status	2026 - LEARNER EDITION
Teaching standard	M01 beginner-first teacher loop. Videos are visual companions; the book remains sufficient fallback.
C2B boundary	Bridge foundations may be assumed, but C3 engineering behavior is demonstrated, broken, repaired, transferred and validated.
Brand	Charcoal #1F2937 | Teal #14B8A6 | Soft Blue #3B82F6 | AMOLED #000

Course rule: preserve what arrived -> validate interpretation -> quarantine ambiguity -> publish durably -> commit state last -> prove replay safety. A green job is never sufficient evidence by itself.

Scenario and architecture

A commerce team receives daily order CSV plus return-event JSON. The old workflow replaced one clean file each day. Build the first restart-safe ingestion utility that preserves source evidence and can explain every accepted/quarantined/current record.

```
daily CSV + return JSON
        |
        v
raw landing + manifest
        |
        v
parse / schema / UTC -----> quarantine
        |
        v
accepted staging + current-state logic
        |
        v
reconcile + durable publish
        |
        v
commit (updated_at, order_id) state LAST
```

PROVIDED INPUTS

shop_pulse_2026-08-21.csv; shop_pulse_2026-08-22.csv (compatible extra field + one timezone-naive record); returns_events.json (includes duplicate event); brief.txt; starter project.

NO HIDDEN PREREQUISITES

Everything required was taught in L01-L14. Do not install Airflow, Kafka, Spark or cloud tooling for this Mini-Lab.

Build in stages

STAGE 0 - SMALL PROOF

Before full integration, land one tiny source file, write one manifest, parse one valid row + one quarantine row, and show checkpoint stays unchanged until validation passes.

STAGE 1 - SOURCE EVIDENCE

Immutable landing per source/run with run_id, source object/window, extracted_at_utc, bytes/record count and fingerprint.

STAGE 2 - INTERPRETATION

Normalize trusted timezone-aware timestamps to UTC; quarantine ambiguous naive time unless contract supplies timezone. Define compatible vs unsafe schema changes.

STAGE 3 - STATE/CURRENT RECORDS

Use stable order boundary such as `(updated_at, order_id)`. Handle legitimate updates by key and explain winner/source version. Deduplicate return events by justified identity while preserving raw duplicate evidence.

STAGE 4 - FAILURE / RECOVERY

Inject failure before checkpoint commit. Rerun from unchanged state and prove no accepted records were skipped; repeated processing must be safe.

STAGE 5 - VALIDATION

Reconcile source/accepted/quarantine/unique keys/checkpoint before-after. Add pytest or equivalent state/boundary tests.

Evidence and interview defense

EVIDENCE BUNDLE

• source + config/contracts

• raw landing + manifests/fingerprints

• accepted/quarantine outputs

• checkpoint before/failure/recovery

• tests

• sanitized logs

• reconciliation report

• README: run, assumptions, timezone/schema policy, replay/recovery and limitations

INTERVIEW DEFENSE

Explain architecture in 3-5 minutes. Then defend: why state commits last; why timestamp-only cursor is risky; one schema/time ambiguity; one injected failure; one independent control; what changes at 100x scale.

RUBRIC - NO AVERAGE

Every M1-M6 competency must PASS and critical failures must equal zero. Scores, averages or percentages are not used. A single required FAIL blocks Mini-Lab PASS.

ID	Competency	PASS evidence	Status
M1	Source contract & immutable landing	Raw files + run/window/source metadata + fingerprint preserved.	PASS / FAIL
M2	Parsing, schema & temporal integrity	Declared schema policy; UTC normalization; ambiguous records quarantined with reasons.	PASS / FAIL
M3	Incremental state correctness	Stable tuple boundary; output durable/validated before checkpoint commit.	PASS / FAIL
M4	Replay & recovery	Injected pre-commit failure; rerun proves no skipped records or unintended accepted duplicates.	PASS / FAIL
M5	Testing & reconciliation	Boundary/state tests + independent source/accepted/quarantine/key controls.	PASS / FAIL
M6	Security, observability & handoff	Sanitized logs, no secrets, README/runbook, operator can reproduce/diagnose.	PASS / FAIL

CRITICAL FAILURES

• Raw/source evidence overwritten or modified in place.

• Checkpoint/state advanced before durable validated publication.

• Unresolved schema/time ambiguity silently coerced into accepted data.

• Replay/backfill demonstrably skips or creates unintended accepted duplicates.

• Real credential/token exposed in source, logs or evidence.

• Fabricated runtime/test/result evidence.

• Protected review used before attempt was frozen.

NEXT ROUTE

PASS does not bypass Gate A. After all competencies pass, attempt C3-024A independently. If the Mini-Lab fails, repair only the weak competency and run a changed integration proof.